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13 SEPTEMBER 1963
SUPERSEDING
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MILITARY SPECIFICATION

CALIBRATOR, THERMOMETER

1. SCOPE

1.1 This specification covers one type of calibrator for calibration of Thermometer, Self-Indicating, Bimetallic, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL-W-45205 — Welding, Inert-Gas, Metal-Arc, Aluminum Alloys Readily Weldable for Structures, Excluding Armor.

STANDARDS

MILITARY

MIL-STD-8 — Dimensioning and Tolerancing.

MIL-STD-9 — Screw Threads Conventions and Methods of Specifying.

MIL-STD-10 — Surface Roughness, Waviness and Lay.

JAN-STD-19 — Welding Symbols.

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-171 — Finishing of Metal and Wood Surfaces.

DRAWINGS

ORDNANCE CORPS

F8032313 — Calibrator, Thermometer.

PUBLICATIONS

ORDNANCE CORPS

MIL-P-14232-

P8032313 — Packaging Data Sheet for Calibrator, Thermometer.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall conform to the material specification(s) referenced on the detail drawings of Drawing F8032313.

3.2 **Dimensions.** Dimensions of parts and assemblies shall conform to the applicable drawings. Interpretation of dimensions, including tolerances and geometrical symbols, shall conform to Standard MIL-STD-8.

3.3 **Construction.**

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3.3.1 *Assembly.* Parts and components shall be positioned and attached as depicted on applicable drawings. Approved source parts shall have met Government approval prior to assembly.

3.3.2 *Threads.* Threads shall conform to the applicable drawings and Standard MIL-STD-9.

3.3.3 *Surface roughness.* Interpretation of surface roughness symbols shall be as defined in Standard MIL-STD-10.

3.3.4 *Welding.* Welding of aluminum alloy parts shall conform to the requirements of Specification MIL-S-45205 Class B. Weld symbols on drawings shall be interpreted as defined in Standard JAN-STD-19. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedure and test specimens.

3.4 Protective coatings.

3.4.1 *Inorganic finishes.* Inorganic finishes shall conform to the finish number(s) of Standard MIL-STD-171 referenced on the applicable drawings.

3.4.2 *Painting.* Unless otherwise specified on the drawings, painting of the Thermometer Calibrator Assembly shall conform to MIL-STD-171, System 21.5. Color shall be Olive Drab as specified by the procuring activity (see 6.2).

3.5 Marking. Marking, including application of part numbers and instructional marking shall conform to the applicable drawings and Standard MIL-STD-130.

3.6 Performance and product characteristics.

3.6.1 *Watertightness.* The Thermometer Calibrator Assembly with an internal air pressure of 2.0 ± 0.5 pounds per square inch (p.s.i.) applied shall show no evidence of air bubbles indicating leakage when submerged in water to a depth of one to two inches below the water level for a period of one minute.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 *Material.* Material used in manufacture of the Calibrator Assembly shall be tested in accordance with the material specification referenced on the part drawings.

4.2.2 *Welding.* The contractor shall obtain Government approval of his welding procedure and test specimens prior to production welding. Inspection and tests of welding shall be in accordance with Specification MIL-W-45205 Class B (see 3.3.4).

4.2.3 *Protective coatings.* Inspection and tests of protective coatings, including painting shall be in accordance with Standard MIL-STD-171.

4.2.4 *Commercial parts.* The Government shall be furnished a statement of findings by the contractor, showing results of examination and tests of commercial parts.

4.3 Sampling for lot acceptance.

4.3.1 *Inspection lot.* All calibrators made under essentially the same conditions shall be considered a lot for purposes of inspection.

4.3.2 *Sampling for examination.* A random sample of calibrators shall be selected from each lot offered to the Government in accordance with Standard MIL-STD-105 at inspection level II. The acceptable quality level shall be 1.5 percent defective.

4.3.3 *Sampling for test.* A random sample of calibrators shall be selected from each lot offered to the Government in accordance with Standard MIL-STD-105 at inspection level L7.

4.4 Examination.

4.4.1 *Complete assemblies.* Each of the samples selected in accordance with 4.3.2 shall be

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examined to verify compliance with this specification. Examination shall be conducted in accordance with Table I. Any completed assembly in the sample containing one or more defects shall be rejected and if the number of defects in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE I. *Classification of defects in accordance with Standard MIL-STD-105*

Categories	Defects	Inspection method
Critical	None defined	
Major	AQL = 1.5 percent	
101	Any part missing or wrong type.	Visual
102	Any part not properly assembled or secured.	Visual
103	Missing welds or visible defects of welding.	Visual
104	Painting defects including runs, sags, blisters, and area of thin film or no film.	Visual

4.4.2 *Packaging and packing.* Examination of preservation, packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

4.5 Test procedure.

4.5.1 *Watertightness.* Each of the calibrator samples selected in accordance with 4.3.3 shall be tested for watertightness in accordance with 4.5.1.1. Failure of any calibrator in the sample to pass this test shall reject the lot.

4.5.1.1 *Method of test.* Each sample shall be tested in the following manner:

(a) All openings of the calibrator shall be sealed (for watertightness) except one National Pipe Thread (NPT) hole, which shall be used for connection of an air hose.

(b) Connect the air hose to this opening and submerge calibrator in water at ambient temperature. Depth of submersion shall be one to two inches below the water level.

(c) Apply air pressure of 2.0 ± 0.5 psi into calibrator and maintain pressure for a minimum period of one minute to determine conformance to 3.6.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. Preservation, packaging, packing and marking of the Thermometer Calibrator shall be in accordance with Packaging Data Sheet MIL-P-14232-P8032313.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of applicable documents.

(c) Place of inspection and tests.

(d) Level of protection required.

6.2 Color. Color Chip No. X24087 (Semi-Gloss, Olive-Drab) of Federal Standard 595 should be furnished the contractor as a color sample for painting.

Custodian:
Army—WC

Preparing activity:
Army—WC
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